

“CloudJet” Steel Structure Intelligent Climbing Inspection Robot System Manual

Updated May 1, 2026

1. Product Overview

1.1 Product Name: “CloudJet” Steel Structure Intelligent Climbing Inspection Robot System

1.2 Product Positioning and Application: Designed for steel infrastructure such as industrial plants, bridges, tower shells, and storage tanks – or components thereof – this system provides intelligent non-destructive testing equipment, systems, and supporting services. It replaces high-risk manual work at heights to achieve automated, high-precision detection of surface and internal damage in steel structures.

1.3 Core Value: Replaces manual high-altitude operations with intelligent technology to reduce safety risks; inspection costs are significantly reduced as sensors only need to be deployed at each measurement point on the surface being tested, eliminating the need for data transmission equipment; leveraging intelligent “frequency-sweep & vision” dual-mode inspection technology, it achieves high-precision identification of surface and internal damage, promptly detects structural safety hazards, and ensures the stable operation of infrastructure.



2. System Components

2.1 Robotic Vehicle Unit: A magnetically attached signal acquisition unit capable of adhering to and climbing both flat and curved surfaces; equipped with a wireless data transmission module to perform data acquisition tasks.

2.2 Sensors: Sensitive components for inspecting steel structure surfaces; they require no power supply when not in data-collection mode and can be deployed on the surface being inspected for extended periods.

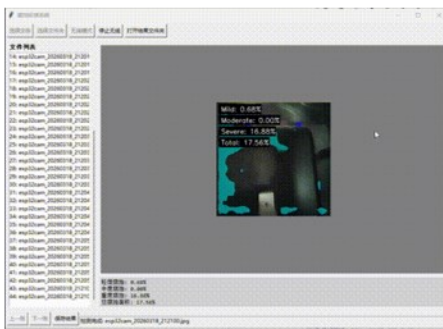
2.3 Remote Control and Data Platform: Control software available on mobile devices and computers enables remote control of the equipment and real-time transmission of inspection footage; the remote cloud monitoring platform facilitates the synchronous upload and viewing of inspection data, supporting subsequent data analysis.



Robot



Sensor



Remote Control and Monitor



Impedance Monitoring Dashboard

“CloudJet” Steel Structure Intelligent Inspection Robot Supporting Equipment and System

3. Key Technical Specifications

3.1 Mobility Parameters:

- (1) Chassis travel speed: 2,400 mm/minute;
- (2) Minimum curved surface diameter for climbing: 50 cm;
- (3) Continuous operating time: ≥ 4 h;

3.2 Inspection Performance Parameters:

- (1) Surface damage assessment resolution: $\leq 0.5 \text{ mm}^2$;
- (2) Internal damage monitoring resolution: $\leq 6.5 \mu\text{m}$;
- (3) Single-point acquisition time: approximately 30 seconds (default value for the scanning frequency range; adjustable);

4. Principle of Operation

4.1 Overview of the System's Overall Operating Principle

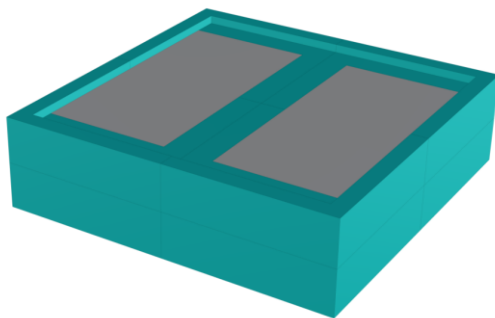
This system adopts a design approach that separates the sensing unit (sensor) from the data acquisition unit (data acquisition and transmission module), consisting of two core components: a piezoelectric sensor and a magnetically-attached mobile data acquisition vehicle. The piezoelectric sensor is permanently mounted on the surface of the steel structure; during non-data-acquisition periods, it requires no power supply or wiring. During data acquisition, the magnetically attached mobile acquisition unit climbs to the measurement point via magnetic adhesion, establishes a temporary electrical connection with the sensor unit, and performs sweep signal acquisition and wireless transmission. Finally, signal analysis is used to identify and quantitatively assess damage to the steel structure.

4.2 Sensor

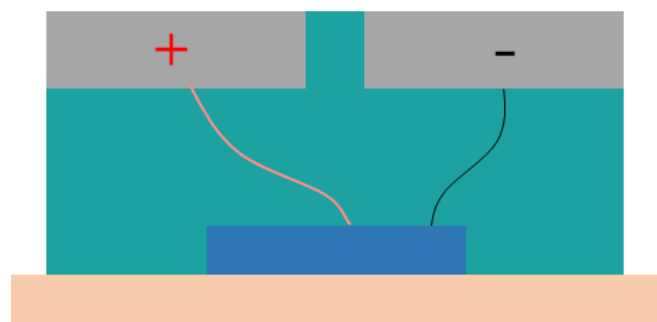
4.2.1 Sensor Structure. The piezoelectric sensor unit features a fully sealed enclosure, primarily comprising: a metal protective housing, a sealing layer, a piezoelectric transducer, a coupling interface, and an exposed corrosion-resistant electrical interface. The electrode surface of the piezoelectric transducer is sealed within the housing, ensuring water resistance, corrosion resistance, and resistance to aging, enabling long-term use in harsh outdoor environments.

4.2.2 Coupling Mechanism. The bottom of the sensor features a coupling surface that adheres tightly to the steel structure using a coupling agent, ensuring stable transmission of vibration and stress and enabling the piezoelectric transducer to deform in unison with the surface being measured.

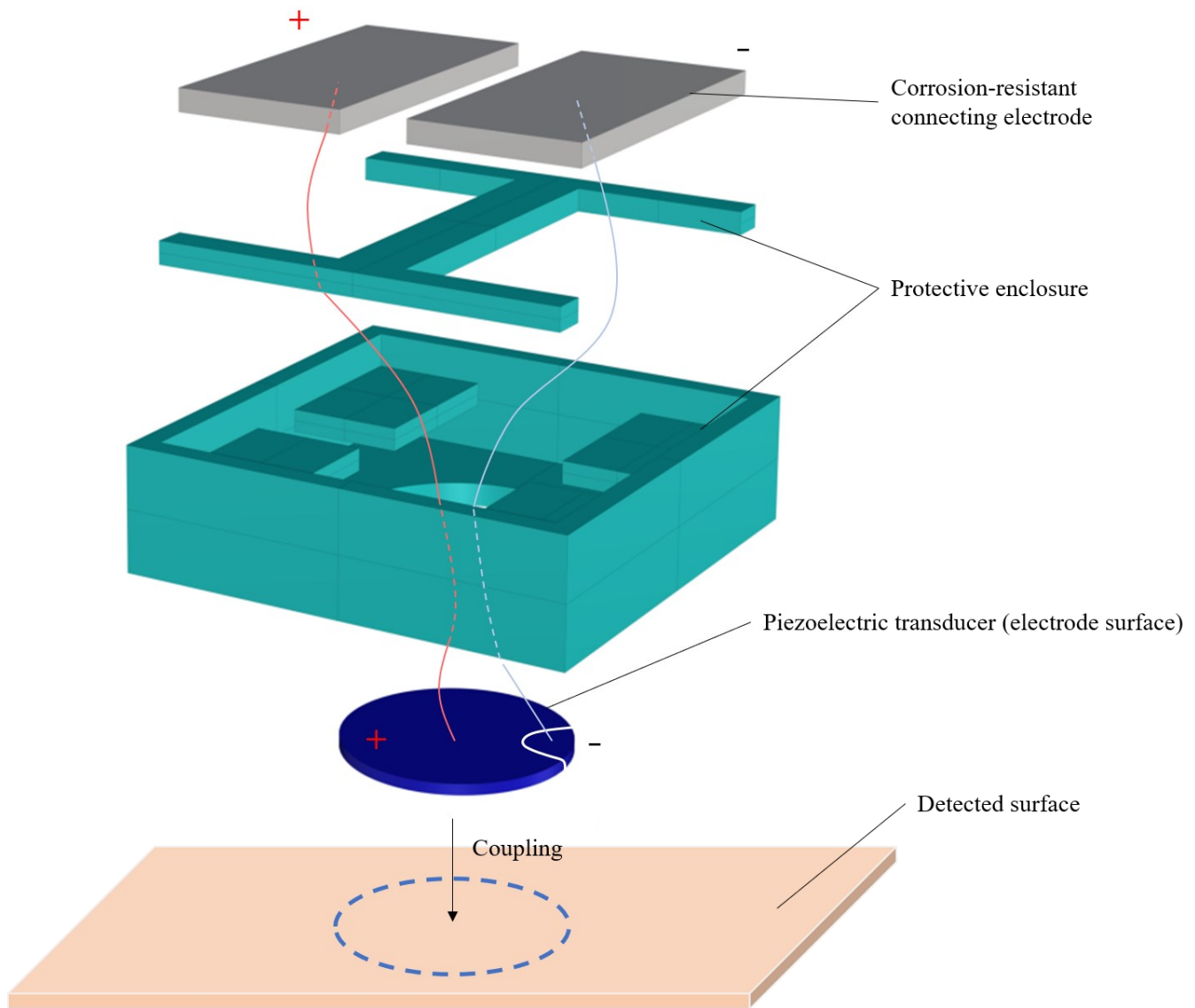
4.2.3 Electrical Interface and Signal Output. The top of the sensor features an exposed, corrosion-resistant electrical interface connected to the positive and negative terminals of the internal piezoelectric transducer. This interface is used for temporary electrical connection to the data acquisition vehicle to facilitate the input of excitation signals and the output of response signals. The sensor does not require power when not in use.



Sensor – External



View Sensor – Side View

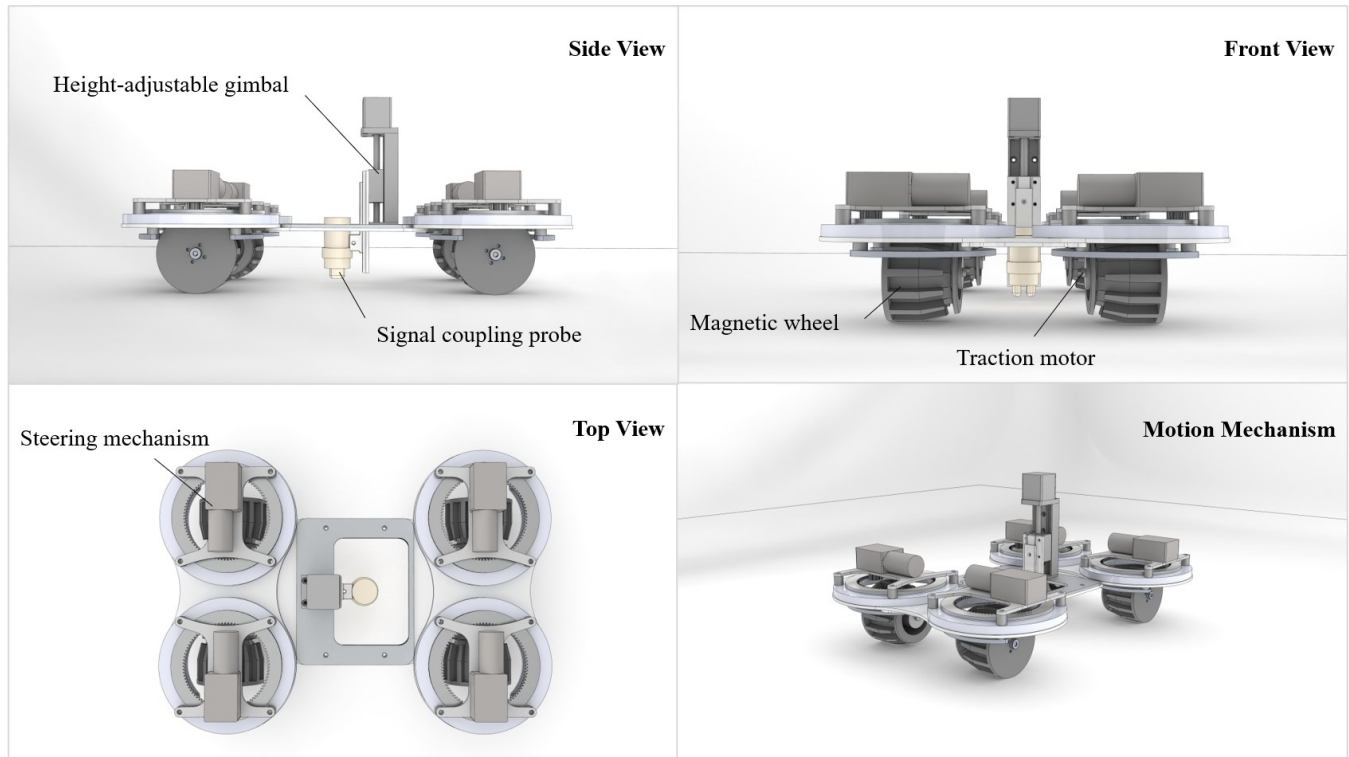


Sensor – Structural Schematic

4.3 Magnetically Attached Mobile Data Collection Unit

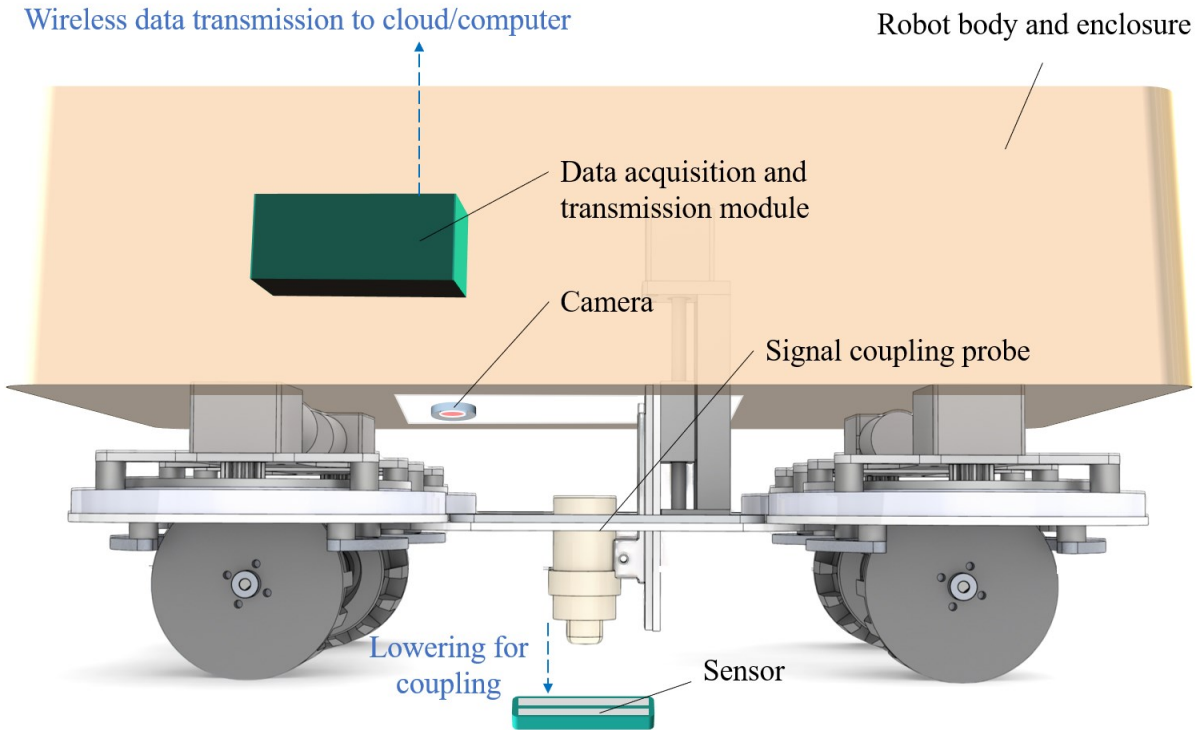
4.3.1 Climbing and Mobility Mechanism. The underside of the unit features curved-surface-adaptive permanent magnet wheels that securely adhere to steel structures via magnetic attraction, accommodating various surface geometries such as flat, positive-curvature, and negative-curvature surfaces. The wheels are equipped with independent traction motors and steering mechanisms, enabling omnidirectional movement and navigation across complex steel structures.

4.3.2 Lifting and Docking Mechanism. The unit is equipped with a lifting gimbal and a rail mechanism that drives the signal docking probe to move vertically, enabling the connection and disconnection of the sensor’s electrical interface.



Robot – Motion Mechanism Schematic

4.3.3 Piezoelectric Sweep Signal Acquisition. The vehicle integrates a piezoelectric sweep data acquisition and transmission module, capable of outputting sweep excitation signals within the 1 to 100 kHz range or specific sub-ranges. When data acquisition is required, the docking probe descends, and the vehicle's data acquisition module establishes an electrical connection with the sensor electrodes via the probe's electrical interface. The sweep signal is applied to the piezoelectric sensing unit through the docking probe's electrical interface, and the feedback electrical signals are simultaneously acquired.



*This figure illustrates the core structural components; secondary components such as the battery and wiring are not shown.

Robot – Core Structure Diagram

4.3.4 Data Transmission. The acquired piezoelectric signals and image data captured by the camera are uploaded in real time via a wireless data transmission module to a remote cloud-based monitoring platform on a computer for subsequent viewing, storage, and analysis.

4.4 Principles of Signal Analysis and Damage Identification

After the acquired sweep signal is processed by the platform, the system identifies and quantitatively analyzes the extent of corrosion, cracks, and damage in steel structures by comparing changes in characteristics such as amplitude, frequency, and phase between the original reference signal and the measured signal. The system supports real-time display of results during the inspection process, automatically saves all image and signal records, and allows for playback at any point and reproduction of results.

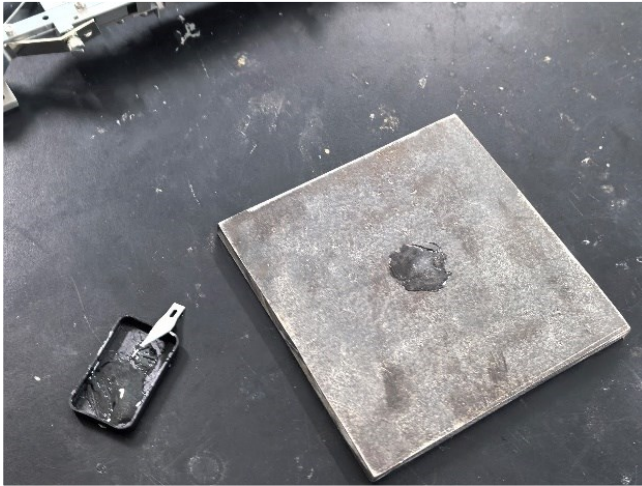
5. Usage Process

5.1 Sensor Deployment

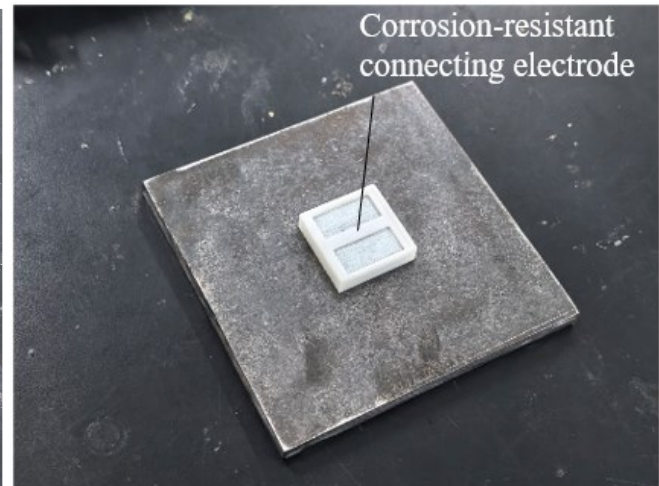
5.1.1 Determine the sensor deployment locations, clean the surface of the measurement areas, and remove contaminants such as dust, oil, and rust to ensure the surface is smooth and clean, thereby achieving optimal sensor coupling and detection accuracy.

5.1.2 Apply an appropriate amount of coupling agent to the measurement point.

5.1.3 Gently press the sensor onto the coupling agent until it adheres tightly to the steel structure’s surface. Then, let it sit for 24 hours to allow the coupling agent to cure; once cured, sensor deployment is complete.



Sensor – Applying Coupling Agent



Sensor – Placing the Sensor

5.2 Equipment Connection and Robot Positioning

5.2.1 Turn on the power to the robot, the remote control unit, and the data analysis unit. Follow the on-screen instructions to connect all three devices to the same local area network (LAN). Open the remote control software on your smartphone or computer, follow the operating instructions to connect the software to the robot, and check the communication signal strength to ensure a stable connection without lag or disconnections.

5.2.2 Once the connection is complete, use the control software to view the robot’s status and check the operational status of the vision module and frequency-sweep detection module to confirm that all equipment is functioning normally.

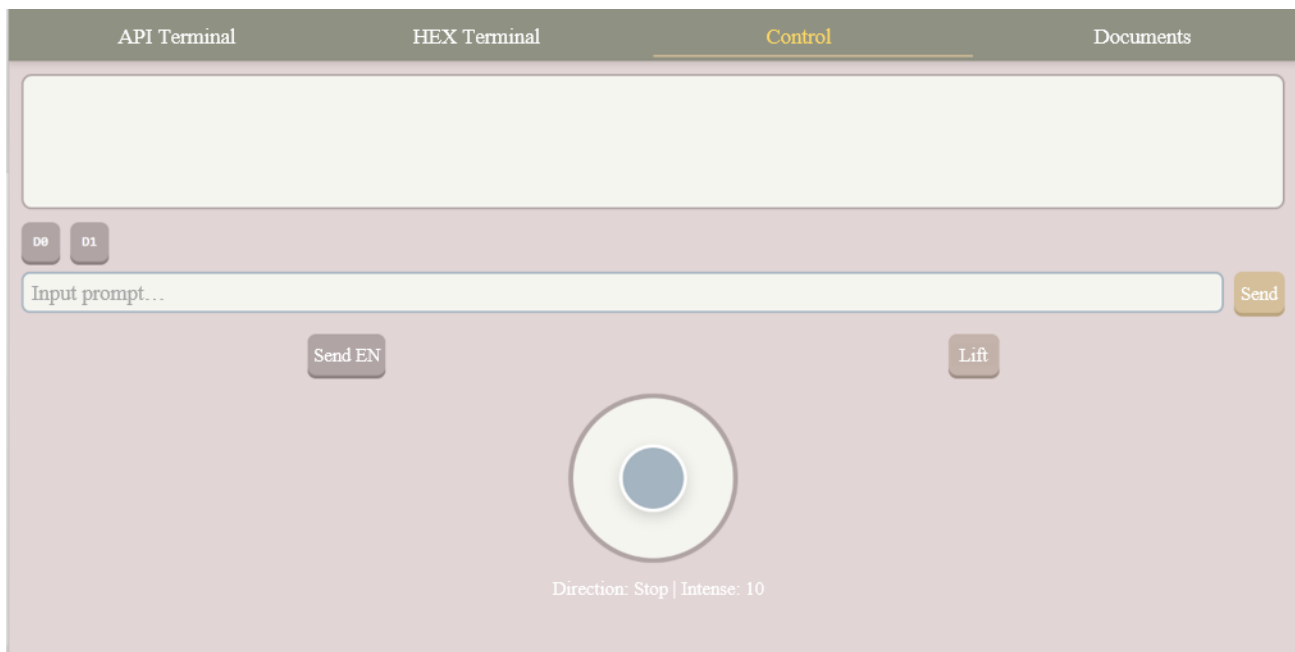
5.2.3 Gently place the robotic vehicle against the starting point of the steel structure inspection, ensuring that all wheels make good contact with the surface of the steel structure, thereby achieving magnetic adhesion.



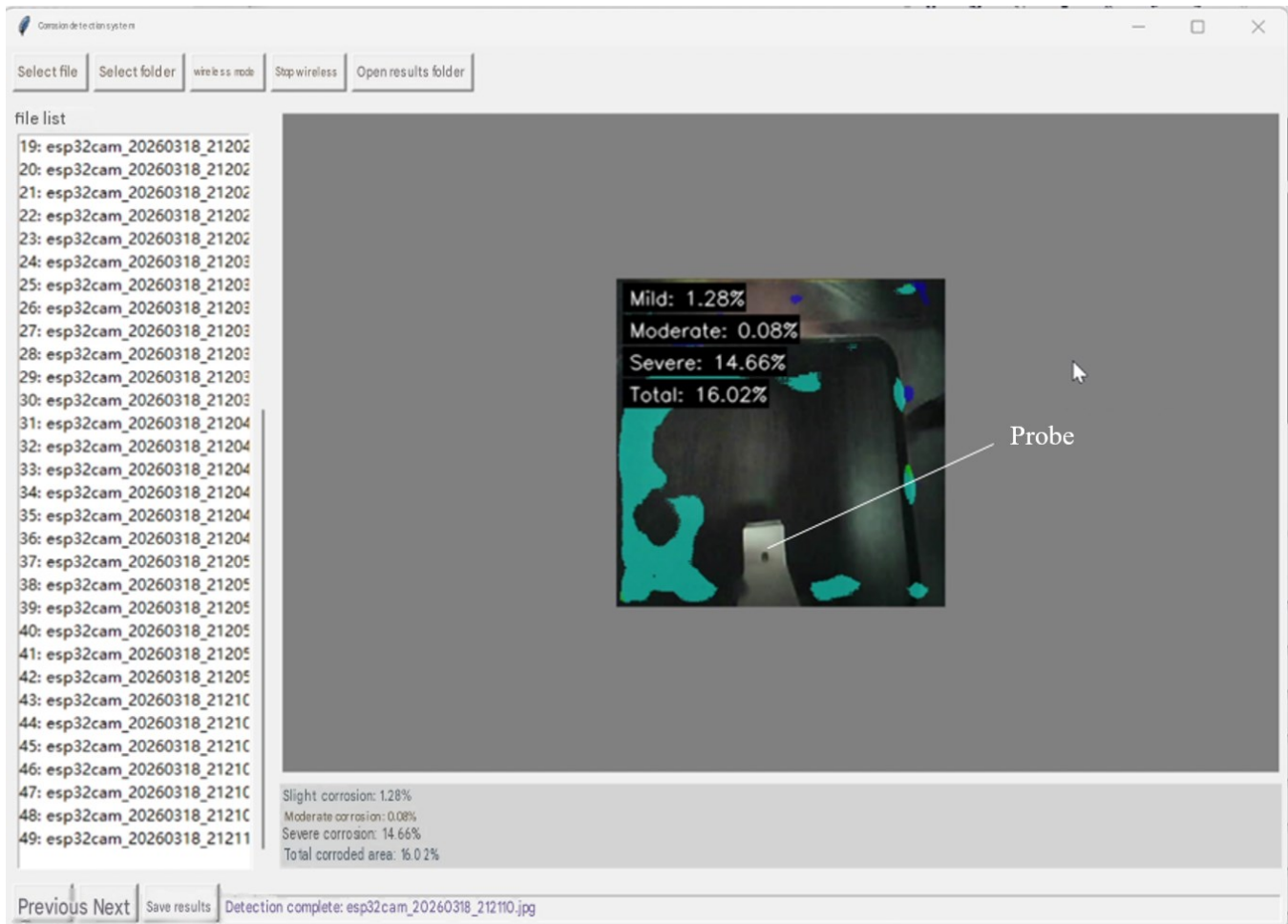
Robot – Climbing Inspection

5.3 Data Collection and Damage Assessment

5.3.1 Operators use remote control software on a computer to maneuver the robot from the control station. Based on the sensor locations, and by visually monitoring the real-time footage transmitted by the camera, they guide the robot so that the bottom docking probe is positioned directly above the first sensor.



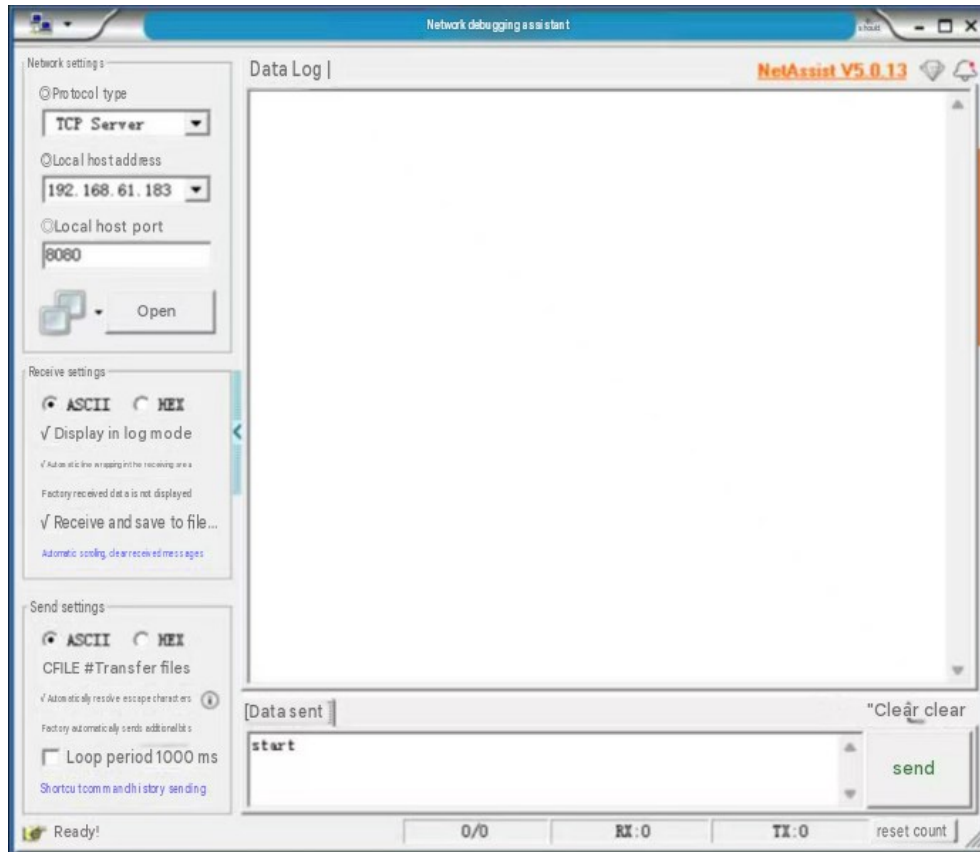
Control Software – Remote Control Interface



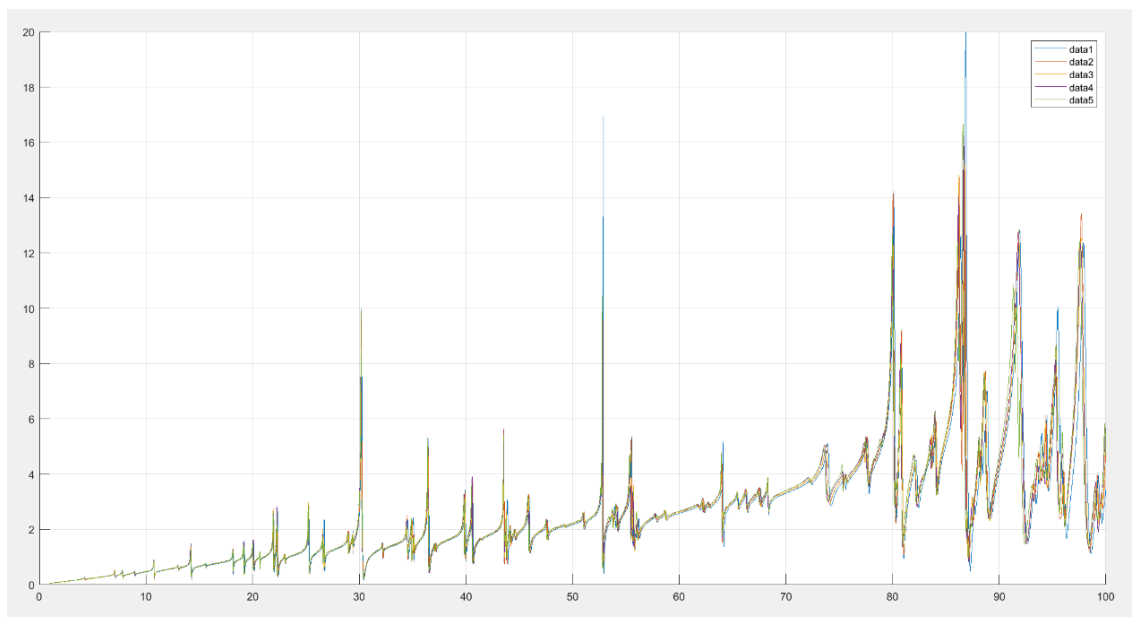
Control Software – Real-Time Camera Feed

5.3.2 Click “Lower” on the remote control interface; the docking probe will automatically descend and automatically complete the docking with the sensor.

5.3.3 Once docking is complete, click “Start” in the control software to initiate data acquisition and set parameters (such as scan frequency range and step size; default values will be used if not specified). The robot then begins collecting swept-frequency data from the steel structure surface corresponding to that sensor and outputs quantified internal damage detection results (such as corrosion rate and crack dimensions).



Control Software - Piezoelectric Sweep Data Acquisition Interface



Control Software – Sweep Data Visualization Output

5.3.4 Once data acquisition for a single sensor is complete, click “Lift” on the remote control; the docking probe will automatically rise and retract, disconnecting from the sensor.

5.3.5 Manually maneuver the robot to the next sensor location and repeat the docking and data acquisition steps described above until data acquisition is complete for all deployed sensors.

5.3.6 During data acquisition, monitor the data transmission status in the control software in real time. If data interruptions or acquisition anomalies occur, immediately stop the operation, inspect the robot, sensors, and connection status, and resume data acquisition only after troubleshooting the issue.

5.4 Data Review and Analysis

5.4.1 Throughout the data collection process, the system displays inspection results in real time. After collection is complete, visual data and sweep signal data can be reviewed via the remote cloud monitoring platform on a computer. The system automatically stores all image and signal records from the inspection area, supporting on-demand playback and result reproduction at any point.

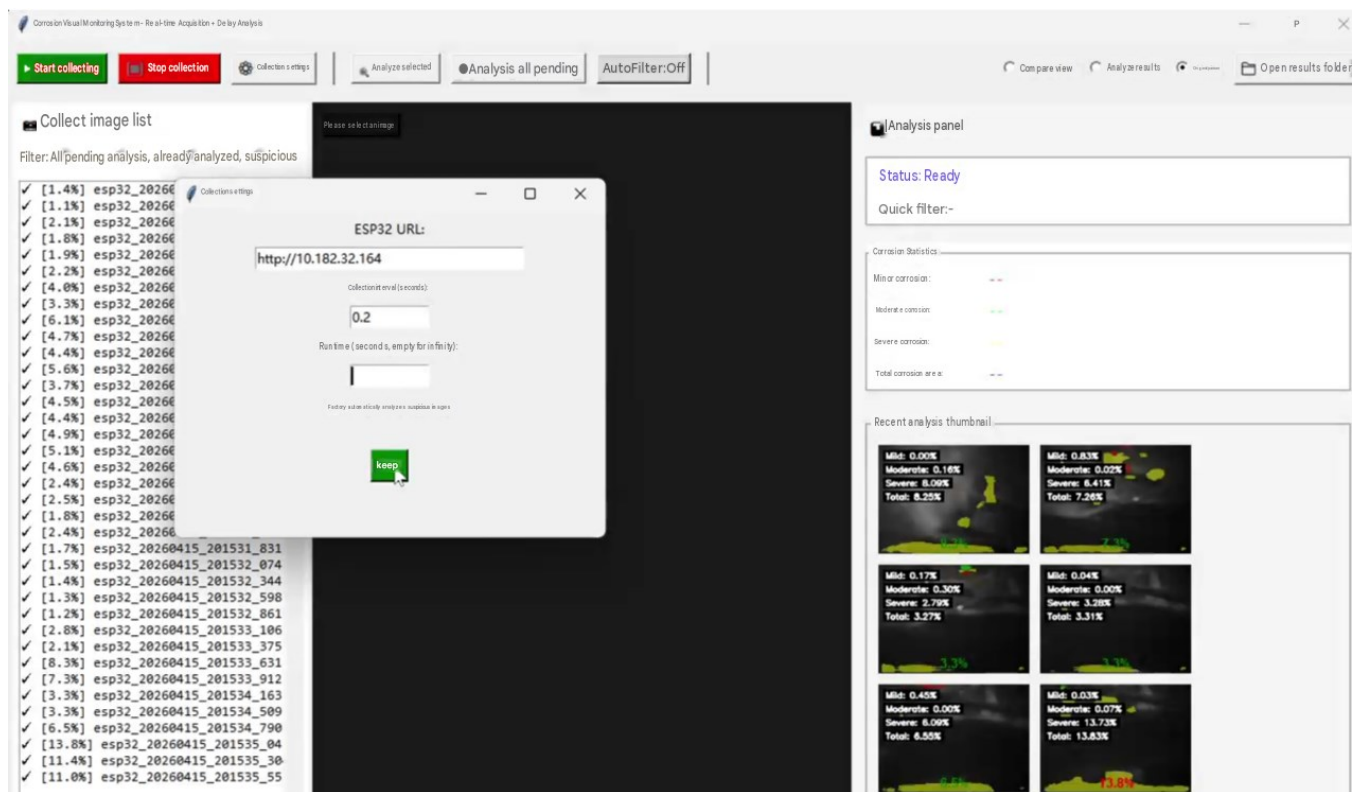


Image Reproduction and Analysis

5.4.2 Upon activating the analysis function, the system automatically processes the collected data to reproduce dual-mode inspection results: surface defect identification based on the vision module and quantitative analysis of internal damage based on the sweep frequency module, thereby generating preliminary overall inspection results.

5.4.3 Operators conduct a preliminary review of the analysis results to verify the accuracy of defect identification and the precision of damage localization. Data and analysis results can be exported for subsequent structural health assessments and potential hazard investigations.

5.5 Task Completion and Equipment Retrieval

5.5.1 Once data analysis is complete, carefully remove the robot from the steel structure surface and place it in the designated storage area.

5.5.2 In the control software, disable data acquisition, analysis, and other functions; disconnect the robot from the computer; and turn off the robot’s power.

5.5.3 Perform basic cleaning of the robot’s surface, removing dust and debris from the probe and magnetic attachment points.

6. Notes

6.1 The parameters and performance metrics described in this manual are based on test results obtained in a laboratory environment. They are provided for reference only and do not represent the final specifications of the product.

6.2 This product is an early prototype; its detection efficiency, stability, and environmental adaptability have not yet been fully verified, and it is not yet capable of industrial-scale batch operations.

6.3 Product features, structure, specifications, and usage procedures will be continuously optimized and updated; subsequent versions may change without prior notice.

6.4 At this stage, this device is limited to R&D testing and technical validation and must not be used directly in formal scenarios such as engineering acceptance or statutory testing.

6.5 All test data and analysis results are provided for R&D reference only and do not constitute final testing conclusions.